CS471 – Lesson 6 In-Class Worksheet

Informed Search (Greedy & A*)

Part 1: Guided In-Class Worksheet (Instructor-Led)

Objective: Understand heuristics and trace Informed Search algorithms.

1. Core Concepts

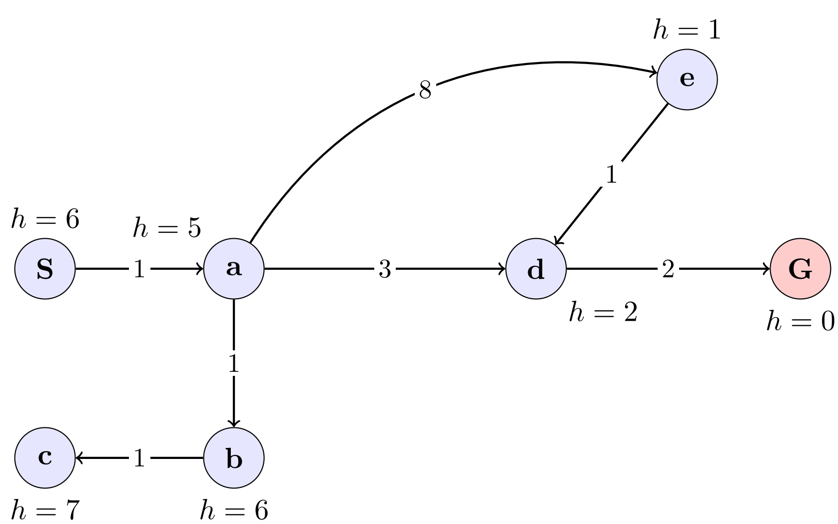
1. **Heuristic ($h(n)$):**

2. **Greedy Search Strategy:**

3. **A* Search Strategy:**

2. Algorithm Tracing

Use the following state space graph to trace both Greedy Search and A* Search. The true step costs are located on the arrows. The heuristic estimates (h) are located outside each node.



A. Greedy Search Trace ($f(n) = h(n)$)

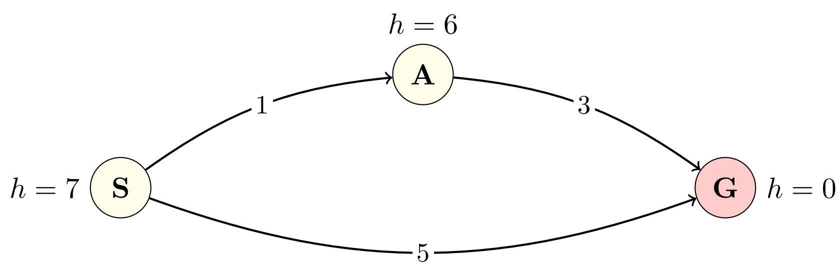
- **Final Path:** ____________________
- **Total True Cost:** __________

B. A* Search Trace ($f(n) = g(n) + h(n)$)

- **Final Path:** ____________________
- **Total True Cost:** __________

3. Admissibility and Optimality

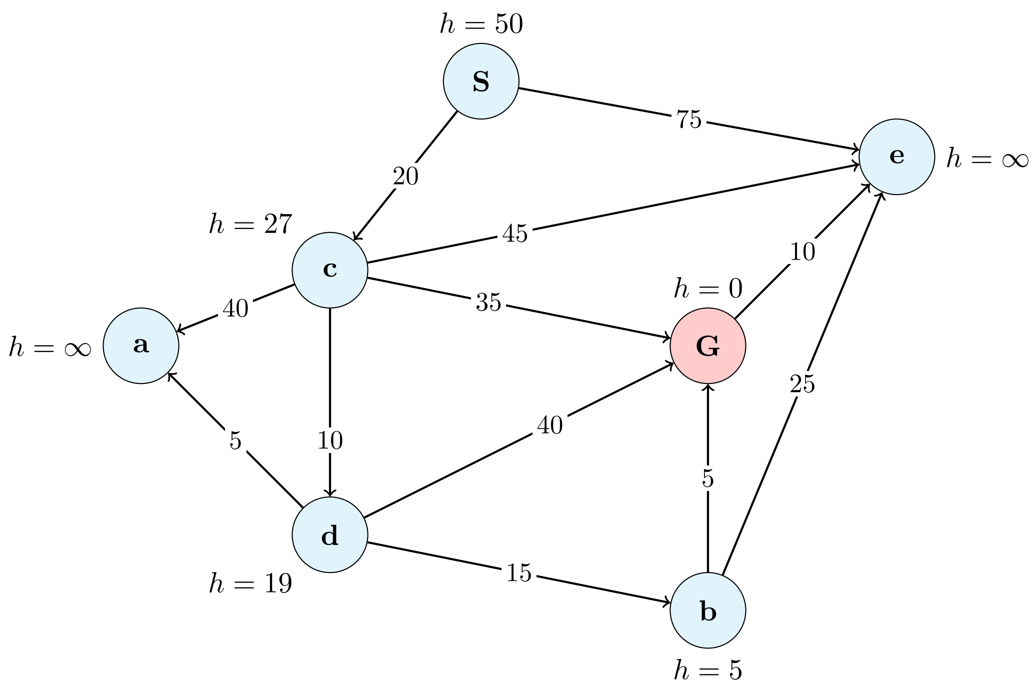
Evaluate the following graph. Is the heuristic admissible? Why did A* fail to find the optimal path?



Part 2: Student Independent Practice

Objective: Independently trace Greedy Search and A* Search on a complex state space.

Use the graph below. The true step costs are on the arrows. The heuristic estimate (h) to the Goal (G) is provided outside each node.



1. Greedy Search Trace ($f(n) = h(n)$)

Trace the path chosen by Greedy Search. Show your frontier priorities at each step.

- **Final Path:** ____________________
- **Total True Cost:** __________

2. A* Search Trace ($f(n) = g(n) + h(n)$)

Trace the path chosen by A* Search. Show your mathematical calculations for $f(n)$ in the frontier.

- **Final Path:** ____________________
- **Total True Cost:** __________